DigiStore License Verification

Sample Python Tkinter Calculator Application — Developer & Testing Guide

Summary: This sample application demonstrates end-to-end integration with the DigiStore Software Licensing API. It is packaged as a single, self-contained Python file (calculator.py) built with Tkinter. It demonstrates: key activation, seat limit enforcement (5/5 seats used), automatic local caching across app restarts, and an in-app **Reset App** workflow that deactivates the device seat on the server.

1. Core Architectural Requirements

- **Single Python File:** The entire application lives inside sample app/calculator.py with zero external GUI dependencies (Tkinter is built into Python).
- **Gumroad-Style Drop-In Snippet:** Contains a self-contained DigiStoreLicense class highlighted with visual banners that developers can copy & paste into ANY software.
- **License Prompt on First Launch:** When an unactivated instance starts, the user is presented with a Software License Verification dialog.
- **Session Persistence (No Repeated Prompts):** Once verified, activation metadata is saved to license_config.json. Subsequent launches load the calculator immediately.
- **Reset App (Seat Release):** Clicking **Reset App** sends a deactivation request (POST /api/licenses/decrement) to the server, releases the seat, deletes the local cache, and returns to the license prompt.
- **5/5 Seat Limit Handling:** If all available activations are consumed, the app explicitly displays **'All keys used'** and blocks activation.

2. The Drop-in SDK Snippet (Copy-Paste Ready)

Just like Gumroad, protecting your own application takes only **3 lines of code** using the highlighted DigiStoreLicense class at the top of calculator.py:

```
# 1. Initialize licensing client
lic = DigiStoreLicense(api_url="http://localhost:3002")

# 2. Check if already activated (persisted across restarts)
if lic.is_activated():
    start_application()
else:
    # 3. Prompt user and activate
    res = lic.activate("DIGI-XXXX-XXXX-XXXX")
    if res["valid"]: start_application()
    elif res["all_keys_used"]: show_error("All keys used!")
```

3. Quick Start Guide

Ensure the DigiStore backend server is running, then launch the application from the terminal:

```
cd "c:\Coding\CLG\FY\finalyr\sample app"
python calculator.py
```

Default Configuration:

- **API Server:** http://localhost:3002 (configurable directly in the activation UI)
- **License Cache:** sample app/license_config.json (created upon successful activation)

4. Step-by-Step Testing Scenarios

Scenario	Action	Expected Result
1. First Launch	Run python calculator.py	Prompts for license key. Calculator is locked.
2. Valid Key Activation	Enter valid DIGI-XXXX-XXXX-XXXX and click Activate.	Calls /api/licenses/verify. Increments server seat count by 1. Stores license_config.json. Calculator opens.
3. App Restart (Persistence)	Close window, then run python calculator.py again.	**Does NOT ask for key again.** Loads calculator instantly with active seat pill displayed.
4. Seat Limit (5/5 Used)	Click **Reset App** ■ → choose **Simulate New Device**. Re-enter key until 5 activations are claimed.	Displays error: **'All keys used (5/5 activations reached)'**.
5. Free Seat / Reset	Click **Reset App** ■ → choose **Release Seat on Server**.	Calls /api/licenses/decrement to free 1 seat on DigiStore, clears local config, and returns to key prompt.
6. Live Sync on Launch	Restart app when server seats change.	Silently queries incrementUses=false to keep top bar keys counter in live sync.

5. Ready-to-Test License Keys from Local Database

The following real license keys are active in your local database for immediate testing:

License Key	Product Name	Initial Seats	Status
DIGI-34A9-4601-6945	Python Test App	1 / 5 Active	Ready to activate (4 seats free)
DIGI-7494-E793-DC67	Python Test App	2 / 5 Active	Ready to activate (3 seats free)
DIGI-8D96-ABBA-74AF	Bento Grid Layouts	0 / 5 Active	Fresh key (5 seats free)

6. DigiStore License API Reference

Endpoint 1: POST /api/licenses/verify

Verifies key validity and increments active seat count upon machine activation.

```
Request Body:
{ "licenseKey": "DIGI-34A9-4601-6945", "incrementUses": true }

Success Response (HTTP 200):
{ "success": true, "valid": true, "uses": 2, "maxUses": 5, "key": "DIGI-34A9-4601-6945", "product": { "id":
"...", "name": "Python Test App" }, "message": "License verified successfully (2/5 seats active)" }

Limit Exceeded Response (HTTP 400 - 5/5 Seats):
{ "success": false, "valid": false, "uses": 5, "maxUses": 5, "message": "Activation limit exceeded: 5 of 5
seats in use. Deactivate an existing device first." }
```

Endpoint 2: POST /api/licenses/decrement

Invoked when a user uninstalls software or clicks **Reset App** to free a seat.

```
Request Body:
{ "licenseKey": "DIGI-34A9-4601-6945" }

Success Response (HTTP 200):
{ "success": true, "uses": 1, "maxUses": 5, "message": "Device deactivated. Available seats: 4/5" }
```

7. Local Storage Format (license_config.json)

```
Upon successful activation, calculator.py creates license_config.json in the same folder:
{ "license_key": "DIGI-34A9-4601-6945", "product_name": "Python Test App", "product_id":
"cmvi7trx70001h82k0t0rdt7j", "uses": 2, "max_uses": 5, "api_url": "http://localhost:3002", "activated_at":
"2026-10-25T14:30:00Z" }
```

When **Reset App** is clicked, this file is automatically deleted from disk.